

ChancaDEM: A Crusher-Comminution Studio Coupling Whiten, Evertsson and Bond, with a Physics-Faithful Learned Surrogate

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CAOS open-research programme · CAOS_ChancaDEM, crusher comminution studio

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_ChancaDEM

Abstract. A rock crusher’s product size distribution, throughput and power are governed by well-established comminution models, but those models are scattered across the literature and rarely coupled in one runnable tool. ChancaDEM is an interactive studio that couples three cited models, a Whiten classification-breakage population balance for the product gradation, an Evertsson flow-capacity model for the throughput hump, and Bond’s law for the power, and lets a user set the machine (cone, jaw or gyratory), eccentric speed, throw, closed-side setting and feed, and see the product form and understand why. On top of the physics engine, a learned surrogate accelerates it: a multilayer-perceptron trained on the population-balance engine reproduces every output, the size percentiles, the passing fractions, the throughput and the power, at a coefficient of determination between 0.980 and 0.998 and a mean absolute percentage error of 2.7% to 10.3%, on a leakage-safe independent Latin-hypercube test draw, and, importantly, it preserves the physical monotonicity of product size against closed-side setting rather than merely fitting the numbers. Over seventeen cases the studio reproduces the textbook comminution behaviour, coarse high-throughput gyratory products through to fine cone products, and the classical Bond trade that a finer product costs more specific energy. The studio couples *calibrated models*, not a plant; the surrogate emulates the *engine* and not a real operation; and the three-dimensional chamber view is a kinematic animation, *not* a discrete-element solve (industrial DEM is infeasible in a browser, and replaying precomputed DEM traces is left to future work). Every number and figure can be regenerated from the released artifacts.

Index Terms. Comminution, crushing, population balance, Whiten model, Bond’s law, cone crusher, surrogate model, machine learning, mineral processing, reproducibility.

I. INTRODUCTION

COMMINUTION, crushing and grinding ore to liberate its minerals, is the largest energy consumer in a mine, and predicting how a crusher turns a feed into a product is a mature modelling problem [6]. The product size distribution follows from a classification-breakage population balance in the tradition of Whiten [1], with the breakage described by

the JKMRC energy-fineness (t_{10}) relation and an appearance function [2]; the throughput follows from the machine’s flow capacity, modelled for cone crushers by Evertsson [4], [5]; and the power follows from Bond’s third theory [3]. Each of these is standard, but a student or an engineer wanting to see how they interact, how eccentric speed, throw and closed-side setting jointly shape the product, gradation, throughput and power, has to assemble them from scattered sources. ChancaDEM assembles them into one runnable studio and adds a learned surrogate that makes the coupled engine instant.

The report describes that studio and, in particular, measures how faithfully the learned surrogate reproduces the physics engine. Faithfulness is not just goodness-of-fit: a surrogate that fits the training numbers but violates the monotonic dependence of product size on closed-side setting would be worse than useless in a teaching or what-if tool, because it would teach the wrong physics. So the surrogate is checked both on accuracy, against a leakage-safe held-out design, and on physical consistency. The contribution is a coupled, surrogate-accelerated comminution studio, with the accuracy and the consistency of each part reported, and with an explicit statement of what the tool is (a coupling of calibrated models) and is not (a plant, or a discrete-element solve, despite the name).

II. THE COUPLED COMMUNITION ENGINE

The engine couples three models, all evaluated in closed form so the whole thing runs live. The **product gradation** comes from the Whiten classification-breakage population balance [1]: for a feed vector $\mathbf{f}$ (mass in size classes), a classification (selection) matrix $\mathbf{C}$ giving the probability a particle in each class is broken, and a breakage matrix $\mathbf{B}$ giving the daughter-size distribution, the product is

$$\mathbf{p} = (\mathbf{I} - \mathbf{C})(\mathbf{I} - \mathbf{B}\mathbf{C})^{-1}\mathbf{f}. \quad (1)$$

The breakage matrix $\mathbf{B}$ is built from the JKMRC t_{10} energy-fineness curve (how much fine material a given breakage energy produces) and an appearance function [2], so the gradation responds physically to the machine setting. The **throughput** follows the Evertsson flow-capacity model [4], [5], which produces the characteristic unimodal capacity hump

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Interactive studio and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ChancaDEM; the studio is available at <https://chancadem.fasl-work.com>.

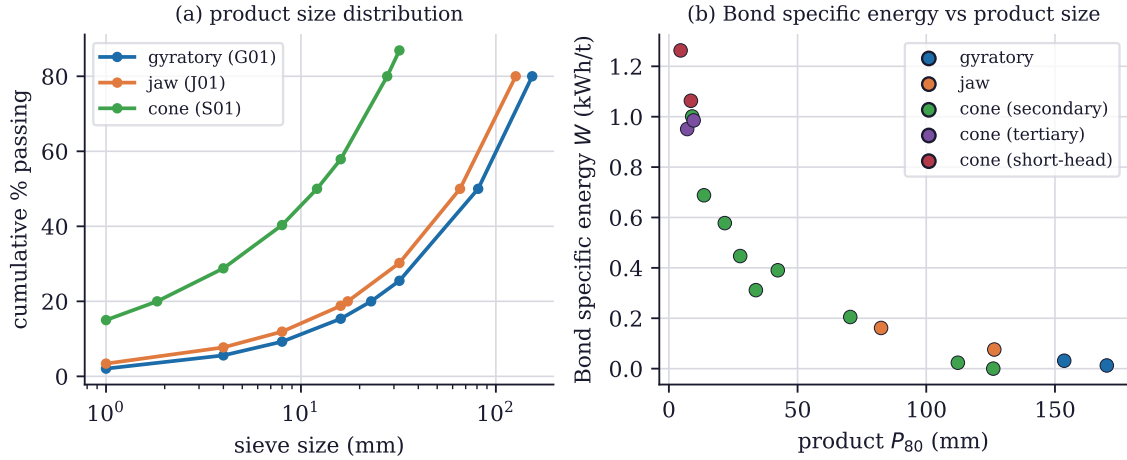


Fig. 1. The coupled engine’s response over seventeen cases. (a) Product size distributions (cumulative percent passing against sieve size) for a gyratory, a jaw and a cone case: the gyratory delivers a coarse high-tonnage product, the jaw an intermediate one, and the cone a fine product, the size-reduction ladder of a crushing circuit. (b) Bond specific energy W , from which the engine computes the power draw ($P = WQ$), against product P_{80} for all seventeen cases, coloured by machine: the finer the product, the more energy per tonne, the classical Bond trade, from about 0.01–0.03 kWh/t for the coarse gyratory products to about 1.3 kWh/t for the finest (short-head cone) product. The two secondary-cone points near zero (P_{80} of 112 and 126 mm) are control settings with the closed side wider than the feed F_{80} , which pass the rock almost unbroken.

against eccentric speed (too slow starves the chamber, too fast chokes it). The **power** follows Bond’s third theory of comminution [3], the specific energy scaling with the difference of the inverse square roots of product and feed sizes. The three are coupled: the same setting that shifts the gradation shifts the capacity and the power, so the studio shows the real trade-offs a crusher operator faces, not three independent dials.

III. THE CRUSHER RESPONSE

Figure 1 shows that the coupled engine reproduces textbook comminution behaviour across seventeen cases spanning the three machine types. The product size distributions (Fig. 1a) form the size-reduction ladder of a real circuit: the gyratory crusher produces a coarse product (P_{80} around 150 mm) at high throughput (thousands of tonnes per hour), the jaw an intermediate product, and the cone a fine product (P_{80} under 30 mm) at lower throughput. The Bond specific energy (Fig. 1b), which the engine computes from the ore work index and the feed and product sizes and multiplies by the throughput to obtain the power draw, makes the classical Bond trade visible: across the cases, finer products cost more energy per tonne, from about 0.01–0.03 kWh/t for the coarse gyratory products to about 1.3 kWh/t for the finest short-head cone product, with the primary crushers at the low end (at most 0.16 kWh/t) and the two short-head cones at the high end (1.06 and 1.26 kWh/t). None of this is surprising: a studio that couples the cited models should reproduce the behaviour the models are known to produce, and it does, which is what licenses using it to teach and to explore.

IV. THE LEARNED SURROGATE

The physics engine is cheap but not instant, and for live what-if exploration the studio runs a learned surrogate, a multilayer perceptron trained on the population-balance engine to map machine settings and feed to all ten outputs. Figure 2

TABLE I
SURROGATE ACCURACY PER OUTPUT AGAINST THE
POPULATION-BALANCE ENGINE (LEAKAGE-SAFE HELD-OUT DRAW,
 $n_{\text{TEST}} = 2196$). ALL OUTPUTS $R^2 \geq 0.980$; THE FINE P_{20} TAIL HAS THE
LARGEST PERCENTAGE ERROR.

Output	R^2	MAPE (%)
P_{80} (product 80% size)	0.995	4.5
P_{50}	0.995	5.0
P_{20}	0.983	10.3
% passing 32 mm	0.997	2.7
% passing 16 mm	0.995	5.1
% passing 4 mm	0.992	6.6
throughput (tph)	0.998	7.6
power (kW)	0.980	6.4

and Table I report its accuracy on a leakage-safe test set: the held-out data are an independent second Latin-hypercube draw (a different seed), not a random split of the training rows, so there is no near-duplicate leakage between train and test. On that test the surrogate reproduces every output at a coefficient of determination between 0.980 (the power) and 0.998 (the throughput), with 0.997 and 0.995 for the coarse passing fractions (32 and 16 mm), and a mean absolute percentage error from 2.7% to 10.3%; the error is largest on P_{20} , the fine tail of the product, which is intrinsically the hardest quantity to predict. Crucially, the surrogate is checked for *physical consistency*, not just fit: it preserves the monotone increase of product P_{80} with closed-side setting, so it will not teach a user the wrong direction of a control. A surrogate that is both accurate to a few percent and monotone in the right variables is a faithful accelerator of the engine; the consistency check is therefore reported alongside the error.

V. DISCUSSION AND SCOPE

ChancaDEM is a coupling of established comminution models made interactive, and a demonstration that a learned

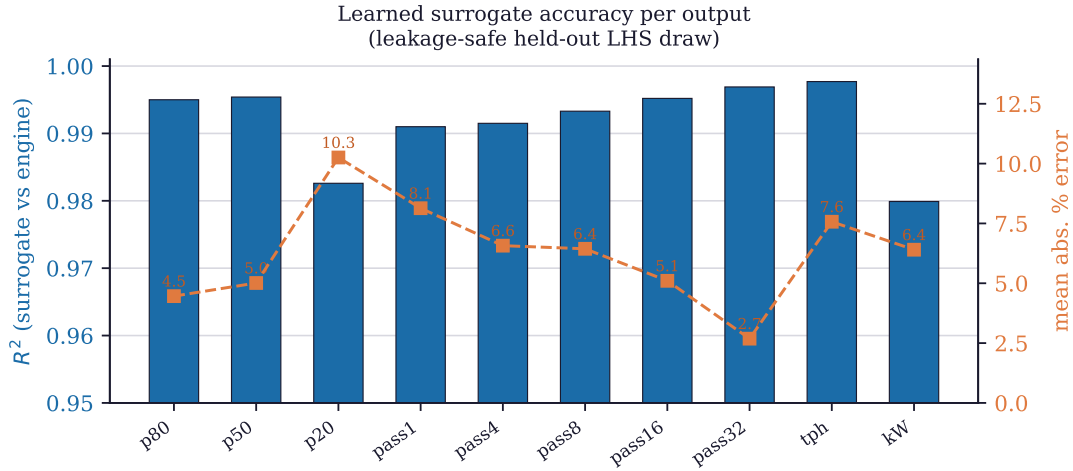


Fig. 2. The learned surrogate against the population-balance engine, per output, on a leakage-safe held-out Latin-hypercube draw (an independent second sampling, not a row split of the training set). The coefficient of determination (bars) is between 0.980 and 0.998 for every output, the product size percentiles, the passing fractions, the throughput and the power; the mean absolute percentage error (line) is 2.7% to 10.3%, worst on the fine P_{20} tail, as expected. The surrogate also preserves the monotone dependence of P_{80} on closed-side setting, so it is physically faithful, not just numerically close.

surrogate can accelerate such a coupling faithfully, to a few percent and with the physics monotonicity intact. The value for a learner or an engineer is seeing the three models interact, that the setting which makes the product finer also moves the capacity hump and raises the power, in one place and in real time, and the surrogate makes that exploration instant. The studio couples *calibrated models*, not a real plant; the parameters are physically reasonable but are not fitted to a specific crusher, so the numbers are the models' behaviour, not a prediction for a given machine. The surrogate emulates the cheap physics *engine*, not a plant, and its accuracy is against that engine. And, despite the name, the three-dimensional chamber view is a *kinematic animation*, geometry and motion driving the engine's gradation, and *not* a discrete-element-method solve; industrial DEM of a crushing chamber is infeasible in a browser, and replaying offline precomputed DEM traces, rather than solving DEM live, is left to future work. A companion denoising-autoencoder anomaly score flags when the surrogate is asked to extrapolate outside its training envelope, so a user is warned rather than silently misled. The studio states the same limits on its methodology pages.

VI. CONCLUSION

ChancaDEM couples the Whiten population balance, the Evertsson capacity model and Bond's power law into one runnable crusher-comminution studio, reproduces the textbook size-reduction ladder and the Bond energy-size trade across seventeen cases, and accelerates the coupled engine with a learned surrogate that is faithful, coefficient of determination 0.980 to 0.998, error a few percent, and monotone in the right variables. The scope is calibrated models rather than a plant, a surrogate of the engine rather than of reality, and a kinematic animation rather than a discrete-element solve; within it, the reproducible coupling and the consistency-checked surrogate are the contribution.

APPENDIX A REPRODUCTION

The repository artifacts under `data/derived/` carry the seventeen cases' engine outputs (`case-results.json`: gradation, throughput, power and the per-nip breakage energy per case; each case's `trace.json` also records its operating point, including the machine type and the ore work index), the surrogate metrics (`surrogate_metrics.json`: per-output R^2 and MAPE on the held-out Latin-hypercube draw, and the P_{80} -monotonicity flag), and the anomaly-detector scaler and threshold. The figure scripts under `manuscripts/comminution-studio/figures/` read those artifacts; Fig. 1b plots the Bond specific energy as the power divided by the throughput. The physics engine and the surrogate are deterministic given their inputs, so re-running reproduces Table I and Figs. 1–2.

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